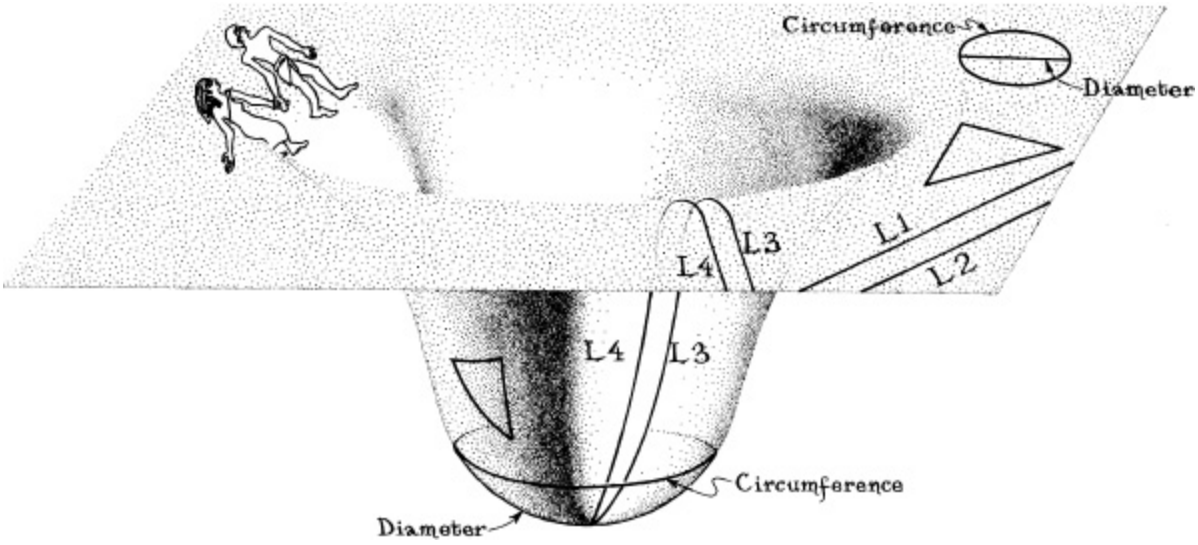
straight lines (geodesics) and then add up the triangle's interior angles, they obtain 180 degrees in the outer, flat region, and more than 180 degrees in the inner, curved region.

Having discovered, by such measurements, that their universe is curved, the 2D beings might begin to speculate about the existence of a three-dimensional space in which their universe resides—in which it is *embedded.* They might give that three-dimensional space the name *hyperspace,* and speculate about its properties; for example, they might presume it to be "flat" in the Euclidean sense that straight, parallel lines in it never cross. You and I have no difficulty visualizing such a hyperspace; it is the three-dimensional space of Figure 3.2, the space of our everyday experience. However, the 2D beings, with their limited two-dimensional experience, would have great difficulty visualizing it. Moreover, there is no way that they could ever learn whether such a hyperspace really exists. They can never get out of their two-dimensional universe and into hyperspace's third dimension, and because they see only by means of light rays that stay always in their universe, they can never see into hyperspace. For them, hyperspace would be entirely hypothetical.



3.2 A two-dimensional universe peopled by 2D beings.

The third dimension of hyperspace has nothing to do with the 2D beings' "time" dimension, which they might also think of as a third dimension. When thinking about hyperspace, the beings would actually have to think in terms of four dimensions: two for the space of their universe, one for its time, and one for the third dimension of hyperspace.